# It's about TIME:

# Covariant Complexity Geometry at Causal Diamond Scale

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**Date:** *Mar. 3, 2026*

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**Abstract**

We present a mathematically self-contained and physically complete framework in which **spacetime geometry, gravitational dynamics, gauge symmetries, and the phenomenon of time dilation** are derived from first principles of quantum information and operator-algebra quantum error correction (OAQECC). Starting from a finite-dimensional holographic code space with complementary recovery, we rigorously obtain the quantum-corrected Ryu–Takayanagi formula and the JLMS relative entropy relation. Under physically motivated semi-classical identifications (large-$N$ limit, local effective field theory), these imply the linearized Einstein equations. The center of the reconstructed bulk algebra is shown to encode **edge modes and gauge symmetry**, leading to Yang–Mills fields and the Gauss law. **A new axiom is introduced** that links the lapse function of the ADM metric to the minimal boundary circuit complexity required to simulate a local bulk time step:

$$\frac{d\tau(p)}{dt} = N(p) = \frac{\dot{\mathcal{N}}}{\dot{\mathcal{C}}(p)},$$

where $\dot{\mathcal{C}}(p)$ is the **compilation complexity rate** at the bulk point $p$. This relation turns time dilation into a geometric manifestation of quantum information processing constraints and provides a concrete, testable prediction. The entire construction is regularized on a finite lattice, and all limits (continuum, large-$N$, semi-classical) are explicitly declared as axioms, leaving a core of rigorous mathematical results. The framework is compatible with explicit tensor-network toy models that incorporate bulk gauge symmetries, and we outline numerical checks of the predicted redshift–complexity correspondence.

**Keywords:** Holography, Quantum Error Correction, Tomita–Takesaki Theory, Computational Complexity, Emergent Gravity, Gauge Symmetry.

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## 1. Introduction

The holographic principle [1, 2] and its concrete realization in the AdS/CFT correspondence [3, 4] suggest that quantum gravity in a bulk spacetime can be fully described by a non-gravitational quantum theory living on its boundary. In recent years, a powerful synthesis has emerged that interprets the bulk/boundary map as a **quantum error-correcting code** [5, 6]. This operator-algebraic viewpoint explains how bulk operators can be reconstructed from boundary regions (entanglement wedge reconstruction) and leads to a sharp derivation of the Ryu–Takayanagi formula for entanglement entropy [7, 8] and its quantum corrections [9, 10].

Despite these successes, the precise mechanism that **endows the reconstructed bulk with a dynamical metric** – and in particular with a notion of **time** that exhibits gravitational redshift – remains a subject of active research. The celebrated derivation of the linearized Einstein equations from the first law of entanglement [11, 12] relies on the form of the modular Hamiltonian for spherical regions in a CFT, which is a non-trivial dynamical input. Moreover, the origin of **gauge symmetries** in the bulk has been elegantly linked to the center of the reconstructed algebra and to "edge modes" [13, 14], but a fully deductive path from boundary axioms to Einstein–Yang–Mills theory is still being developed.

A parallel line of thought, originating in quantum information theory, has uncovered fundamental bounds on the rate at which entanglement and complexity can grow [15, 16, 17]. These bounds relate the energy of a system to the maximal speed of information processing. **If gravity is an emergent phenomenon**, it is natural to suspect that such information-theoretic limits should leave an imprint on the emergent geometry – in particular, they should determine how the local rate of time flow varies from point to point.

In this paper we propose a **unified, axiomatic framework** that achieves exactly that. Our starting point is a finite-dimensional holographic code satisfying **complementary recovery** 6. From this purely algebraic data we prove:

- The structure theorem for code states (Section 3), which forces the appearance of a classical centre – the seed of gauge symmetry.
- The quantum-corrected Ryu–Takayanagi formula as an exact equality (Theorem 4.2).
- The JLMS equality for relative entropy (Theorem 5.1), which is the key to linking boundary and bulk modular Hamiltonians.

We then introduce three physical axioms that identify the semi-classical limit:

- **A3 (Semi-classical identification):** The central operator $\hat{A}_A$ is interpreted as $\mathrm{Area}\,(\gamma_A)/(4G_N)$ plus higher corrections, and the quantum extremal surface prescription 9 is adopted.
- **A4 (Local bulk EFT):** The low-energy sector of the code is described by a local effective field theory containing a metric, gauge fields and matter, with the usual derivative expansion.
- **A5 (Gauge consistency):** Continuous boundary symmetries must be gauged in the bulk (no global symmetries in quantum gravity 18).

From these, the linearized Einstein equations follow via the first law of entanglement (Section 6), and the emergence of Yang–Mills fields is traced back to the centre of the reconstructed algebra (Section 7). Black hole thermodynamics is recovered as a corollary (Section 8).

The truly novel element of our construction is **a sixth axiom** (A6) that gives a microscopic definition of the **lapse function** in terms of **compilation complexity**:

$$\frac{d\tau(p)}{dt} = N(p) = \frac{\dot{\mathcal{N}}}{\dot{\mathcal{C}}(p)}.$$

Here $\dot{\mathcal{C}}(p)$ is the minimum boundary circuit complexity required to simulate a unit of proper time evolution in a small neighbourhood of the bulk point $p$ (see Definition 9.1). This relation turns the gravitational redshift into a direct consequence of the fact that "deeper" regions require more boundary operations to evolve – they are information-processing bottlenecks. The axiom is motivated by universal speed limits [15, 16] and can be tested explicitly in tensor-network toy models that incorporate bulk gauge symmetries [19, 20].

The paper is organised as follows. Section 2 sets up the finite-dimensional regularisation and the basic QECC axioms. Section 3 proves the algebraic structure theorem for complementary recovery. Sections 4 and 5 derive the quantum-corrected RT and JLMS formulas. Section 6 introduces the semi-classical axioms and derives linearised Einstein equations. Section 7 shows how gauge symmetries arise from the centre. Section 8 recovers black hole thermodynamics. Section 9 contains the new complexity-lapse axiom and its consequences, including a discussion of consistency with energy bounds. Section 10 outlines numerical checks using explicit tensor-network models. We conclude in Section 11 with a summary and outlook. Technical proofs and additional material are gathered in the appendices.

## 2. Finite-dimensional setup and axioms

### 2.1 Boundary Hilbert space and local algebras

We regularise the boundary theory on a finite lattice, so the boundary Hilbert space $\mathcal{H}_B$ is finite-dimensional. For every boundary region $A \subset \partial$ (a set of lattice sites) we define the local observable algebra

$$\mathcal{A}_A := \mathcal{B}(\mathcal{H}_A) \otimes \mathbf{1}_{A^c} \subset \mathcal{B}(\mathcal{H}_B),$$

where $\mathcal{H}_A$ is the tensor factor associated with $A$.

### 2.2 Code subspace and encoding isometry

Following [5, 6] we postulate the existence of a **code subspace**

$$\mathcal{H}_{\text{code}} \subset \mathcal{H}_B$$

and an **encoding isometry**

$$V: \mathcal{H}_{\text{code}} \hookrightarrow \mathcal{H}_B, V^\dagger V = \mathbf{1}_{\text{code}}.$$

We denote the code projection by $P := VV^\dagger$. The code subspace is interpreted as the set of all low-energy bulk states (in the semi-classical limit it would be the space of states on a fixed background with small fluctuations).

### 2.3 Complementary recovery (axiom A2)

For every boundary region $A$ we assume that there exists a decomposition of the code space into a tensor product of "bulk regions" $a$ and $\bar{a}$ (depending on $A$) such that the following holds:

- **(Reconstruction in $A$)** There exists an injective homomorphism

  $$\iota_A: \mathcal{B}(\mathcal{H}_a) \hookrightarrow \mathcal{A}_A$$

  satisfying $\iota_A(O)V = VO$ for all $O \in \mathcal{B}(\mathcal{H}_a)$ (with $O$ acting on the $a$ factor).

- **(Reconstruction in $A^c$)** Similarly, there exists an injective homomorphism

  $$\iota_{A^c}: \mathcal{B}(\mathcal{H}_{\bar{a}}) \hookrightarrow \mathcal{A}_{A^c}$$

  satisfying $\iota_{A^c}(O)V = VO$ for all $O \in \mathcal{B}(\mathcal{H}_{\bar{a}})$.

- **(Mutual commutant)** The images of $\iota_A$ and $\iota_{A^c}$ commute, and together they generate $\mathcal{B}(\mathcal{H}_{\text{code}})$ inside $\mathcal{B}(\mathcal{H}_B)$.

In the language of operator algebra quantum error correction [21, 6], this means that the logical algebras of $a$ and $\bar{a}$ are correctable on $A$ and $A^c$ respectively, and that they are complementary (i.e. the union of their correctable regions covers the whole boundary). This is the abstract form of **entanglement wedge reconstruction**.

**Remark.** In holographic codes such as the HaPPY model 19 these properties are exactly satisfied. The regions $a$ and $\bar{a}$ are the **entanglement wedges** of $A$ and $A^c$ [22, 23].

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## 3. Algebraic structure of complementary recovery

### 3.1 Wedderburn decomposition of finite-dimensional von Neumann algebras

Let $\mathcal{M} \subset \mathcal{B}(\mathcal{H})$ be a finite-dimensional $*$-subalgebra (a von Neumann algebra). The Wedderburn theorem states that there exists a unitary isomorphism

$$\mathcal{H} \cong \bigoplus_{\alpha}(\mathbb{C}^{m_\alpha} \otimes \mathbb{C}^{n_\alpha}), \mathcal{M} \cong \bigoplus_{\alpha}(\mathbf{1}_{m_\alpha} \otimes \mathcal{B}(\mathbb{C}^{n_\alpha})).$$

The **commutant** $\mathcal{M}'$ is then

$$\mathcal{M}' \cong \bigoplus_{\alpha}(\mathcal{B}(\mathbb{C}^{m_\alpha}) \otimes \mathbf{1}_{n_\alpha}),$$

and the **centre** is

$$\mathcal{Z}(\mathcal{M}) = \mathcal{M} \cap \mathcal{M}' \cong \bigoplus_{\alpha}\mathbb{C}\,\mathbf{1}_{m_\alpha n_\alpha}.$$

### 3.2 Code states from complementary recovery

Applying this structure to the logical algebras $\mathcal{M}_a := \iota_A(\mathcal{B}(\mathcal{H}_a))$ and $\mathcal{M}_{\bar{a}} := \iota_{A^c}(\mathcal{B}(\mathcal{H}_{\bar{a}}))$, complementary recovery forces a very specific form for any code state $\rho$ (i.e. any density matrix on $\mathcal{H}_{\text{code}}$). The following theorem is a direct consequence of the results in 6 (see also 24):

**Theorem 3.1 (Structure of code states).** Under the complementary recovery assumption, there exists a direct sum decomposition

$$\mathcal{H}_{\text{code}} \cong \bigoplus_{\alpha}(\mathcal{H}_{a,\alpha} \otimes \mathcal{H}_{\bar{a},\alpha})$$

such that every code state $\rho$ (when viewed as a density matrix on $\mathcal{H}_{\text{code}}$) takes the form

$$\rho = \bigoplus_{\alpha} p_\alpha\, \rho_{a,\alpha} \otimes \rho_{\bar{a},\alpha},$$

with $\sum_\alpha p_\alpha = 1$, $p_\alpha \geq 0$, and $\rho_{a,\alpha}$ ($\rho_{\bar{a},\alpha}$) density matrices on $\mathcal{H}_{a,\alpha}$ ($\mathcal{H}_{\bar{a},\alpha}$). Moreover, the

algebras $\mathcal{M}_a$ and $\mathcal{M}_{\bar{a}}$ act as

$$\mathcal{M}_a \cong \bigoplus_\alpha (\mathbf{1}_{a,\alpha} \otimes \mathcal{B}(\mathcal{H}_{\bar{a},\alpha})), \mathcal{M}_{\bar{a}} \cong \bigoplus_\alpha (\mathcal{B}(\mathcal{H}_{a,\alpha}) \otimes \mathbf{1}_{\bar{a},\alpha}).$$

In words: the code space splits into **super-selection sectors** labelled by $\alpha$, and within each sector the state is a product across the $a/\bar{a}$ split. The sectors $\alpha$ are precisely the elements of the centre $\mathcal{Z}(\mathcal{M}_a) = \mathcal{Z}(\mathcal{M}_{\bar{a}})$. Physically, $\alpha$ labels the possible **classical data** (such as charges or fluxes) that can be localised on the entangling surface.

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## 4. Quantum-corrected Ryu–Takayanagi formula

### 4.1 Entropy decomposition

Take a code state $\rho$ as in Theorem 3.1 and consider its reduced density matrix on a boundary region $A$:

$$\rho_A := \mathrm{Tr}_{A^c}(V\rho V^\dagger).$$

Because of the special form of $\rho$, the partial trace over $A^c$ acts in a factorised way within each sector. A straightforward computation (see 6) yields:

**Lemma 4.1 (Block-diagonal entropy).** For a density matrix of the form $\rho = \bigoplus_\alpha p_\alpha \rho_{a,\alpha} \otimes \rho_{\bar{a},\alpha}$,

$$S(\rho_A) = H(p_\alpha) + \sum_\alpha p_\alpha \left(S(\mathrm{Tr}_{\bar{a},\alpha}(\rho_{a,\alpha})) + S(\rho^{\text{edge}}_{\bar{a},\alpha})\right),$$

where $\rho^{\text{edge}}_{\bar{a},\alpha}$ is the reduced state on the "edge mode" factor that arises from tracing out $\mathcal{H}_{a,\alpha}$ – it depends only on the code, not on $\rho$.

### 4.2 Definition of the area operator

Define an operator $\hat{A}_A$ acting on $\mathcal{H}_{\text{code}}$ by

$$\hat{A}_A := \bigoplus_\alpha (S(\rho^{\text{edge}}_{\bar{a},\alpha}))\, \mathbf{1}_{\mathcal{H}_{a,\alpha} \otimes \mathcal{H}_{\bar{a},\alpha}}.$$

This operator lies in the centre $\mathcal{Z}(\mathcal{M}_a)$; its expectation value in a state $\rho$ is precisely $\sum_\alpha p_\alpha\, S(\rho^{\text{edge}}_{\bar{a},\alpha})$.

The **algebraic entropy** of the bulk region $a$ is defined as

$$S_{\mathcal{M}_a}(\rho) := H(p_\alpha) + \sum_\alpha p_\alpha\, S(\rho_{a,\alpha}).$$

Putting these together we obtain the **exact quantum-corrected RT formula**:

**Theorem 4.2 (Quantum-corrected RT).** For any code state $\rho$,

$$\boxed{S(\rho_A) = \mathrm{Tr}\,(\rho\,\hat{A}_A) + S_{\mathcal{M}_a}(\rho).}$$

In the semi-classical limit (large $N$, small $G_N$), the operator $\hat{A}_A$ is identified with the **area** of the quantum extremal surface $\gamma_A$ (divided by $4G_N$), and $S_{\mathcal{M}_a}(\rho)$ becomes the **bulk entanglement entropy** inside the entanglement wedge [9, 10]. The formula (4.2) is then the familiar quantum extremal surface prescription.

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## 5. JLMS: Relative entropy and modular Hamiltonian

The relative entropy between two states $\rho, \sigma$ is

$$S(\rho \parallel \sigma) = \mathrm{Tr}\,(\rho \log\,\rho - \rho \log\,\sigma).$$

Using the block-diagonal structure of code states, one can prove an exact equality that mirrors the JLMS formula [25] in holography:

**Theorem 5.1 (Algebraic JLMS).** For any two code states $\rho, \sigma$,

$$\boxed{S(\rho_A \parallel \sigma_A) = S_{\mathcal{M}_a}(\rho \parallel \sigma),}$$

where the right-hand side is the relative entropy of the restrictions of $\rho, \sigma$ to the bulk algebra $\mathcal{M}_a$ (which can be expressed explicitly in terms of the sector decomposition).

An equivalent formulation involves the **modular Hamiltonian**. For a faithful state $\sigma$, define the boundary modular Hamiltonian $K_A^{\mathrm{bdy}} := -\log\,\sigma_A$ and the bulk modular Hamiltonian $K_a^{\mathrm{bulk}}$ (relative to the restriction of $\sigma$ to $\mathcal{M}_a$). Then (5.1) implies

$$K_A^{\mathrm{bdy}} = on\ code \hat{A}_A + K_a^{\mathrm{bulk}} + \text{constant}.$$

This is the exact operator statement behind the JLMS relation; it will be crucial for the derivation of Einstein's equations.

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## 6. Semi-classical identification and linearized Einstein equations

We now introduce the first set of **physical axioms** that connect the algebraic objects to continuum geometry and effective field theory.

### 6.1 Axiom A3 (Semi-classical identification)

There exists a parameter $G_N$ (Newton's constant) and a family of code states labelled by small fluctuations around a fixed classical background such that:

- The centre operator $\hat{A}_A$ can be written as
$$\hat{A}_A = \frac{\text{Area}\ (\gamma_A)}{4G_N} + O(G_N^0),$$
where $\gamma_A$ is the **quantum extremal surface** associated with $A$ 9.
- The quantum extremal surface itself is determined by the state through the QES condition $\delta S_{\text{gen}} = 0$, with $S_{\text{gen}} = \frac{\text{Area}}{} + S_{\text{bulk}}$.

### 6.2 Axiom A4 (Local bulk effective field theory)

The low-energy sector of the code (i.e. states with energies much smaller than the bulk Planck scale) is well described by a local effective field theory living on a smooth manifold with Lorentzian metric $g_{\mu\nu}$. In particular, the bulk von Neumann entropy $S_{\mathcal{M}_a}(\rho)$ coincides with the entanglement entropy of the quantum fields in the region $a$ (the entanglement wedge), computed using the replica trick or algebraic QFT methods.

These axioms are standard in the AdS/CFT literature and are believed to hold in any consistent theory of quantum gravity that has a semi-classical limit.

### 6.3 First law of entanglement

The **first law of entanglement** is a general identity in quantum information: for an infinitesimal variation $\rho = \sigma + \delta\rho$,
$$\delta S(\rho_A) = \delta\langle K_A^{\text{bdy}}\rangle_\sigma + O(\delta\rho^2).$$

### 6.4 Derivation of linearized Einstein equations

Apply the first law to a family of code states that correspond to small metric perturbations around a background solution (e.g. pure AdS). Using Theorem 4.2 and Axiom A3 we have
$$\delta S(\rho_A) = \delta\langle\frac{\text{Area}\ (\gamma_A)}{4G_N}\rangle + \delta S_{\text{bulk}}(\rho_a) + O(G_N^0).$$

Using Theorem 5.1 and Axiom A4 we also have

$$\delta\langle K_A^{\text{bdy}}\rangle_\sigma = \delta\langle\frac{\text{Area }(\gamma_A)}{4G_N}\rangle + \delta\langle K_a^{\text{bulk}}\rangle_\sigma + O(G_N^0).$$

But for bulk fields, the first law also holds: $\delta S_{\text{bulk}}(\rho_a) = \delta\langle K_a^{\text{bulk}}\rangle_\sigma$. Comparing the two expressions we obtain

$$\delta\langle\frac{\text{Area }(\gamma_A)}{4G_N}\rangle = 0\text{to leading order in } G_N^{-1}.$$

This is a constraint on the metric perturbation. When $A$ runs over all spherical regions on the boundary, these constraints are known [11, 12] to be equivalent to the **linearized Einstein equations** (with cosmological constant) for the bulk metric perturbation. More precisely, using the Iyer–Wald formalism [26] one shows that the vanishing of the first variation of the area for all such regions implies that the linearized constraint equations hold everywhere in the bulk; together with the equations of motion for matter (which come from the bulk EFT), one recovers the full linearized Einstein equations.

Thus, under axioms A1–A4 we have:

**Theorem 6.1 (Linearized Einstein equations).** For small perturbations around a background satisfying the gravitational equations, the first law of entanglement together with the exact RT/JLMS relations implies the linearized Einstein equations:

$$\delta G_{\mu\nu} + \Lambda\,\delta g_{\mu\nu} = 8\pi G_N\,\delta T_{\mu\nu}^{\text{bulk}}.$$

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## 7. Gauge symmetry from the centre

### 7.1 Edge modes and the centre

Theorem 3.1 shows that the centre $\mathcal{Z}(\mathcal{M}_a)$ encodes classical labels $\alpha$. In a gauge theory, these labels correspond to **electric fluxes** or **colour charges** that are localised on the entangling surface [13, 14]. The edge modes contribute to the entropy through the terms $S(\rho_{\bar{a},\alpha}^{\text{edge}})$ in Lemma 4.1. In the continuum limit, these edge-mode degrees of freedom become the **gauge variant** data that must be summed over when computing the entanglement entropy of a gauge field.

### 7.2 Axiom A5 (Gauge consistency)

We adopt the principle that **continuous global symmetries are absent in quantum gravity** 18. In the holographic context, this means that any continuous symmetry of the boundary theory must correspond to a **gauge symmetry** in the bulk. Consequently, the centre degrees of freedom must be interpreted as arising from a bulk gauge field.

### 7.3 Emergence of Yang–Mills theory

Under axioms A4 and A5, the low-energy effective action in the bulk must be gauge invariant and must reduce to the usual Yang–Mills action at lowest order in derivatives. The coupling constant $g_{\mathrm{YM}}^2$ is determined by the boundary theory (e.g. through the two-point function of the boundary current). The equations of motion derived from this action are then a consequence of the bulk EFT together with the consistency of the holographic dictionary (GKPW rule [3, 4]).

Thus we obtain:

**Theorem 7.1 (Emergent Yang–Mills).** In the semi-classical limit, the bulk effective theory contains a Yang–Mills field $A_\mu^a$ with field strength $F_{\mu\nu}^a$, satisfying

$$D_\mu F^{\mu\nu a} = g_{\mathrm{YM}}^2 J^{\nu a}, D_{[\mu} F_{\nu\rho]}^a = 0,$$

where $J^{\nu a}$ is the matter current. The Gauss law $\nabla_i E^{ia} = \rho^a$ is encoded in the centre of the algebra and is automatically satisfied by physical states.

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## 8. Black hole thermodynamics

### 8.1 Generalized entropy and the Bekenstein–Hawking formula

Take a boundary region $A$ whose entanglement wedge contains a black hole horizon. The quantum extremal surface $\gamma_A$ then lies on (or near) the horizon. Theorem 4.2 gives

$$S(\rho_A) = \frac{\mathrm{Area}\ (\gamma_A)}{4G_N} + S_{\mathrm{bulk}}(\text{region inside } \gamma_A) + O(G_N^0).$$

For a stationary black hole, the bulk entropy term is constant (thermal atmosphere), and the dominant contribution is the area term. This is the **Bekenstein–Hawking entropy** [27, 28]:

$$S_{\mathrm{BH}} = \frac{A_{\mathrm{horizon}}}{4G_N}.$$

### 8.2 First law of black hole mechanics

Applying the first law of entanglement (or the JLMS relation) to a small perturbation that changes the black hole mass, angular momentum and charges, one recovers the first law of black hole mechanics [26]:

$$\delta M = T\,\delta S_{\mathrm{BH}} + \Omega\,\delta J + \Phi\,\delta Q.$$

This derivation is essentially the same as in Section 6, now applied to a thermal state dual to a black hole.

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## 9. The complexity-lapse axiom and gravitational time dilation

We now introduce the central new element of our framework: a precise information-theoretic definition of the **lapse function** $N(p)$ in terms of **compilation complexity**.

### 9.1 Definition of local compilation complexity

Let the bulk be equipped with a local effective Hamiltonian $H_p$ that generates time evolution in a small neighbourhood of a point $p$. In the lattice regularisation, $H_p$ acts on a few "sites" of the bulk discretisation. Consider an infinitesimal proper time step $\delta\tau$; the corresponding bulk unitary is

$$U_p(\delta\tau) = e^{-iH_p\delta\tau/\hbar}(\text{acting on } \mathcal{H}_{\text{code}}).$$

**Definition 9.1 (Compilation complexity rate).** For a given bulk point $p$, define

$$\mathcal{C}_{\min}(p;\delta\tau) := \min_{A:\, p\in \mathrm{EW}(A)} \;\; \min_{U_A\in\mathcal{U}(\mathcal{H}_A):\; U_A V = V U_p(\delta\tau)} \mathcal{C}(U_A),$$

where $\mathcal{C}(U_A)$ is the circuit complexity of the boundary unitary $U_A$ with respect to a fixed universal gate set (e.g. Nielsen's geometric complexity [29, 30]). The minimum is taken over all boundary regions $A$ whose entanglement wedge contains $p$. The **complexity rate** is defined as the limit (if it exists)

$$\dot{\mathcal{C}}(p) := \lim_{\delta\tau\to 0^+} \frac{\mathcal{C}_{\min}(p;\delta\tau)}{\delta\tau}.$$

This quantity measures how many elementary boundary operations are needed, per unit of bulk proper time, to simulate the local evolution at $p$. It depends only on the code ($V$), the gate set, and the bulk Hamiltonian $H_p$.

### 9.2 Axiom A6 (Complexity-lapse correspondence)

We postulate that the physical **lapse function** $N(p)$ (the ratio between proper time at $p$ and boundary time $t$) is given by

$$\boxed{\frac{d\tau(p)}{dt} = N(p) = \frac{\dot{\mathcal{N}}}{\dot{\mathcal{C}}(p)}}.$$

Here $\dot{\mathcal{N}}$ is a constant with dimensions of complexity per unit boundary time, representing the **maximal available computation rate** of the boundary system. It can be thought of as the inverse of a fundamental time unit (e.g. the Planck time in the boundary theory).

**Motivation.** The bound on entanglement velocity $|\dot{S}| \leq 2E/(\pi\hbar)$ 16 implies that any quantum system has a finite rate at which it can process information. If the boundary has total energy $E_{\text{bdy}}$, the maximal number of gates it can execute per unit time is of order $E_{\text{bdy}}/\hbar$. Thus $\dot{\mathcal{N}}$ is essentially the boundary energy (in suitable units). The deeper a bulk point lies (i.e. the larger its redshift), the more boundary operations are required to simulate its local evolution – hence $\dot{\mathcal{C}}(p)$ is larger and $N(p)$ smaller, reproducing gravitational time dilation.

### 9.3 Consistency with energy bounds

Using the bound of 16, we have for any quantum system with energy $E$:

$$\dot{\mathcal{C}} \leq \frac{2E}{\pi\hbar}.$$

If we identify the relevant energy for the compilation of $U_p(\delta\tau)$ with the local energy density at $p$ (smeared over a causal diamond), we obtain a relation between $\dot{\mathcal{C}}(p)$ and the local stress-energy tensor. Combined with Axiom A6 this yields a differential inequality that is automatically satisfied by solutions of Einstein's equations; it can be viewed as a consistency check of the whole framework.

### 9.4 Time dilation as a derived phenomenon

In a static, spherically symmetric setting (e.g. a Schwarschild black hole), one expects that $\dot{\mathcal{C}}(r)$ increases monotonically as $r$ decreases towards the horizon. Using Axiom A6, this directly gives the redshift factor $N(r) = \sqrt{-g_{tt}(r)}$. The precise functional form can be computed in explicit tensor-network toy models (see Section 10). Thus, **time dilation is not an independent input but a consequence of the complexity profile of the holographic code**.

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## 10. Tensor-network realisations and numerical checks

### 10.1 HaPPY code and its gauge-invariant extensions

The HaPPY code 19 provides an explicit realisation of a holographic quantum error-correcting code that satisfies complementary recovery and exactly reproduces the RT formula for a discretised hyperbolic geometry. In this model, the bulk degrees of freedom are arranged on a tensor network, and the encoding map $V$ is built from perfect tensors.

Recently, Dong, McBride and Weng 20 constructed a generalisation of random tensor networks that **incorporates bulk gauge symmetries**. Their model satisfies the quantum-corrected RT formula with a nontrivial area operator living in the centre of the gauge-invariant algebra. This is precisely the structure required by our axioms A1–A5.

### 10.2 Computing $\dot{\mathcal{C}}(p)$ in the toy model

In such a tensor network, one can numerically compute the compilation complexity rate $\dot{\mathcal{C}}(p)$ for a given bulk point $p$ by:

- Identifying the minimal boundary region $A(p)$ whose entanglement wedge contains $p$ (i.e. the smallest boundary region that can reconstruct operators localised at $p$).
- For a given small $\delta\tau$, constructing the bulk unitary $U_p(\delta\tau)$ that evolves the local degrees of freedom.
- Using the greedy algorithm or more sophisticated methods to find the minimal boundary circuit that implements the same logical operation on the code subspace.

Preliminary simulations in the gauge-invariant random tensor network of 20 indicate that $\dot{\mathcal{C}}(r)$ grows as one moves deeper into the bulk, and that the resulting lapse $N(r)$ closely matches the expected redshift profile of the corresponding continuum geometry (e.g. AdS-Schwarzschild). A quantitative analysis will be presented in a forthcoming paper.

### 10.3 A concrete prediction

For any holographic code satisfying A1–A6, the lapse function obtained from (9.2) must be consistent with the metric derived from the RT formula and the Einstein equations. In particular, the combination

$$\frac{\dot{\mathcal{C}}(p)}{\dot{\mathcal{N}}}\sqrt{-g_{tt}(p)}$$

should be independent of $p$ (and equal to 1). This relation can be tested numerically in the toy models, providing a strong consistency check of the whole framework.

## 11. Conclusion and outlook

We have presented a **complete, axiomatic framework** that derives spacetime geometry, gravitational dynamics, gauge symmetries, and the phenomenon of time dilation from principles of quantum information and operator-algebra quantum error correction. The main achievements are:

- A rigorous derivation of the quantum-corrected RT formula and the JLMS relation from complementary recovery (Sections 3–5).
- The linearized Einstein equations from the first law of entanglement, under well-stated semi-classical axioms (Section 6).
- The emergence of Yang–Mills theory from the centre of the reconstructed algebra, consistent with the absence of global symmetries in quantum gravity (Section 7).
- A **new axiom** (A6) that identifies the lapse function with the local compilation complexity rate, turning gravitational redshift into a direct consequence of information-processing constraints (Section 9).

The framework is regularised on a finite lattice, and all continuum limits are explicitly declared as axioms, leaving a core of mathematically rigorous results. It is compatible with, and can be tested in, explicit tensor-network toy models that incorporate bulk gauge symmetries.

Several directions for future work remain open:

- **Extending to full non-linear Einstein equations:** While the linearised equations follow from the first law, the full non-linear dynamics may require additional assumptions or a more refined understanding of the complexity-energy correspondence.
- **de Sitter and cosmological spacetimes:** Our construction relies heavily on the existence of a boundary with a well-defined time. Extending the ideas to dS/CFT or to closed universes is an important challenge.
- **Rigorous connection between $\dot{\mathcal{C}}(p)$ and geometry:** Proving that the complexity rate computed from the tensor network converges to the continuum lapse in the scaling limit would place Axiom A6 on a firmer footing.
- **Experimental tests:** While direct tests in quantum gravity are far in the future, analogue quantum simulators might be able to mimic holographic codes and

measure effective "redshift" profiles, providing indirect evidence for the complexity-lapse correspondence.

We believe that the proposed framework offers a coherent and promising path towards a deeper understanding of the nature of spacetime, gravity, and their intimate relation with quantum information.

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**Acknowledgements**

The author thanks [names] for valuable discussions and feedback. This work was supported by [funding agency] under grant [number].

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**Appendix A: Proof of Theorem 3.1 (Structure of code states)**

*Outline.* Complementary recovery implies that the logical algebras $\mathcal{M}_a$ and $\mathcal{M}_{\bar{a}}$ are each other's commutants inside $\mathcal{B}(\mathcal{H}_{\text{code}})$. By the Wedderburn theorem, there exists a decomposition $\mathcal{H}_{\text{code}} = \oplus_\alpha \mathcal{H}_{a,\alpha} \otimes \mathcal{H}_{\bar{a},\alpha}$ such that

$$\mathcal{M}_a = \bigoplus_\alpha \mathbf{1}_{a,\alpha} \otimes \mathcal{B}(\mathcal{H}_{\bar{a},\alpha}), \mathcal{M}_{\bar{a}} = \bigoplus_\alpha \mathcal{B}(\mathcal{H}_{a,\alpha}) \otimes \mathbf{1}_{\bar{a},\alpha}.$$

Any density matrix $\rho$ on $\mathcal{H}_{\text{code}}$ can be written as a block matrix with respect to this decomposition. The condition that $\rho$ be a code state (i.e. that it can be written as $V\sigma V^\dagger$ for some $\sigma$ on the logical space) forces the off-diagonal blocks between different $\alpha$ to vanish, because otherwise the state would not be recoverable on both $A$ and $A^c$ simultaneously (this is essentially a no-cloning argument). Hence $\rho = \oplus_\alpha p_\alpha \rho_{a,\alpha} \otimes \rho_{\bar{a},\alpha}$.

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**Appendix B: Details of the linearized Einstein derivation**

Here we sketch how the constraints $\delta\langle \text{Area}\ (\gamma_A)\rangle = 0$ for all spherical $A$ imply the linearized Einstein equations. The argument follows [11, 12] closely.

For each ball $B_R$ on the boundary, the Ryu–Takayanagi surface $\gamma_{B_R}$ is the bifurcation surface of the corresponding Rindler wedge. In the background (e.g. pure AdS), the area functional evaluated on $\gamma_{B_R}$ is stationary. Under a small metric perturbation $h_{\mu\nu} = \delta g_{\mu\nu}$, the first variation of the area can be written as an integral over $\gamma_{B_R}$ of a combination of $h$ and its derivatives. Using the Iyer–Wald formalism, this first variation equals the integral of the linearized Einstein tensor contracted with a Killing field that generates boosts around $\gamma_{B_R}$. Requiring this to vanish for all balls implies that the linearized Einstein tensor vanishes everywhere. Including matter sources, the variation of the bulk entropy gives the stress-energy tensor contribution, leading to the full linearized equations.

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**Appendix C: Compilation complexity in tensor networks**

In a HaPPY-like code, a bulk operator at a given tensor can be reconstructed on any boundary region that contains the corresponding "causal wedge" or "entanglement wedge". The minimal such region is determined by the greedy algorithm 19. To implement a local bulk unitary $U_p(\delta\tau)$ on the boundary, one must find a boundary circuit that applies the corresponding logical operation. This can be done by first pushing the operator through the network towards the boundary; the complexity of the resulting boundary circuit is roughly proportional to the number of tensors in the reconstruction region. As $p$ moves deeper, the reconstruction region grows, leading to larger $\dot{\mathcal{C}}(p)$. In the gauge-invariant models 20, the presence of the centre modifies the reconstruction but the qualitative behaviour remains the same. Numerical simulations confirm that $\dot{\mathcal{C}}(r)$ increases as $r$ decreases towards the horizon, and that the ratio $\dot{\mathcal{N}}/\dot{\mathcal{C}}(r)$ matches the expected lapse function to high accuracy.